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Face Recognition Attendance Using Python

The main purpose of this project is to build a face recognition-based attendance monitoring system for educational institution to enhance and upgrade the current attendance system into more efficient and effective as compared to before. The current old system has a lot of ambiguity that caused inaccurate and inefficient of attendance taking. Many problems arise when the authority is unable to enforce the regulation that exist in the old system. The technology working behind will be the face recognition system. The human face is one of the natural traits that can uniquely identify an individual. Therefore, it is used to trace identity as the possibilities for a face to deviate or being duplicated is low. In this project, face databases will be created to pump data into the recognizer algorithm. Then, during the attendance taking session, faces will be compared against the database to seek for identity. When an individual is identified, its attendance will be taken down automatically saving necessary information into a excel sheet. At the end of the day, the excel sheet containing attendance information regarding all individuals are mailed to the respective faculty.

Software Requirements:

- Windows 10
- Anaconda
- Some Python Packages
- Camera (Web Cam)

Hardware Components:

- Processor – i3
- Hard Disk – 5 GB
- Memory – 1GB RAM

Advantages:

- The software can be used for security purposes in organizations and in secured zones.
- The software stores the faces that are detected and automatically marks attendance.
- The system is convenient and secure for the users.
- It saves their time and efforts.

Disadvantages:

- The system don't recognize properly in poor light so may give false results.
- It can only detect face from a limited distance.

Applications:

- The system can be used for places that require security like bank, military etc.
- It can also be used in houses and society to recognize the outsiders and save their identity.
- The software can used to mark attendance based on face recognition in organizations.